



Empowering the Faculty of Education Students: Applying AI's Potential for Motivating and Enhancing Learning

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Abstract

This study aimed to investigate how AI impacts students' intrinsic motivation and learning experiences in various cultural and academic settings. An analysis of the effects of AI-powered learning tools on intrinsic motivation and learning enhancement was conducted in early 2024. The study recruited 455 students (192 males and 263 females) from Egypt, Saudi Arabia, Spain, and Poland who were studying diverse majors in educational faculties. Using a 5-point Likert scale, the validated instrument included three dimensions with a total of 10 items: intrinsic motivation and AI, AI and enhancing learning, and overall perception of AI in education. Significant differences were found by nationality and major, with Polish students and education technology majors showing the highest motivation levels. As a result of enhancing learning experiences, autonomy, and critical thinking skills, AI tools were found to positively influence motivation. Motivation, however, was not significantly influenced by academic level. Despite ethical concerns, students held a moderately positive view of artificial intelligence in education. Further research on optimal AI implementation and ethical considerations in education is necessary based on these findings.

Keywords Artificial intelligence · Motivation · Enhancing learning · Faculty of education

Introduction

The study of artificial intelligence (AI) is an important part of computer science education. In AI classes, tools that dynamically display and animate complicated concepts can improve the learning experience (Amershi et al., 2005). In higher education, AI-powered technologies may transform teaching and learning (Lim et al.,

2023; Mohamed, 2024). Universities have the option of integrating their educational programs with different platforms and systems. For example, the use of AI-based solutions can improve students' learning by providing creative methods to convey concepts, collecting data from students and teachers, and delivering individualized feedback to students instantly (Alenezi et al., 2023; Hwang & Chen, 2023). Additionally, these technologies can be used to solve problems, navigate, and develop curricula collaboratively. In addition to data analysis and academic assistance, AI-powered tools can be used to enhance learning through games and challenges. Moreover, it can serve as a dynamic assessor and an adaptive indicator of student performance on assignments (Ivanov & Soliman, 2023; Strzelecki, 2024).

Motivation is a key element in the learning process for achieving success (Filgona et al., 2020). Researchers have found that learners' motivation to learn can be affected by their involvement and learning outcomes (Alamer, 2015; Brooker et al., 2018; Huang et al., 2023). The evidence suggests that the learning experience impacts learners' motivation, and this motivation interacts dynamically with their environment (Alamer, 2015). The motivation of students to learn is strongly influenced by their level of engagement in learning. Motivation and results can be improved by direct teaching strategies, demonstrating a relationship between motivation and results. To increase students' intrinsic motivation to learn and enhance their academic achievement, educators should increasingly utilize interactive educational approaches. These approaches may improve students' motivation and academic performance by boosting their involvement in learning (Huang et al., 2023).

Several studies indicate that students' motivation greatly impacts their learning strategies, engagement, perseverance in reaching goals, and cognitive processes (Alamer & Alrabai, 2023; Chiu et al., 2023; Tremblay-Wragg et al., 2021). It is widely recognized that successful learners are people who possess motivation. Researchers have examined how many motivational factors can predict learning results (Alamer & Alrabai, 2023). A student's willingness to utilize and acquire knowledge from AI technologies may depend on how effectively they can be applied in practice. Studying and implementing artificial intelligence in education (AIED) requires a thorough understanding of student motivation in schools since motivation during adolescence is particularly fragile (Bhat et al., 2024). There is a direct relationship between motivation and AIED that needs to be studied. Teachers can provide useful insights into the practical application of AI technologies by recognizing their importance in inspiring students and helping them succeed academically (Huang et al., 2023; Muthmainnah et al., 2022).

Educational environments can benefit from personalized learning through AI technology. In addition to adaptive learning systems, recommendation systems and personalized learning systems are all forms of intelligent tutoring systems. AI tutoring systems can improve students' academic performance (Nemorin et al., 2023; Steenbergen-Hu & Cooper, 2014). The use of artificial intelligence in higher education institutions has numerous benefits, including enhancing students' skills, promoting collaborative learning in higher education, increasing security, and facilitating research (Kuleto et al., 2021; Strzelecki, 2023; Ziemba et al., 2024).

This study aims to explore the research questions in the developing field of AI in education on intrinsic motivation and learning enhancement:

Research Question 1: What is the impact of AI in education on intrinsic motivation and enhancing learning across different majors, levels, and cultures?

Research Question 2: How do AI-powered learning tools affect students' intrinsic motivation?

Research Question 3: To what extent does AI integration enhance learning?

Research Question 4: What are students' overall perceptions of the impact of AI technologies on education?

This study significantly contributes to the current literature because it examines the impact of artificial intelligence on education in four countries: Saudi Arabia, Egypt, Spain, and Poland. A key goal of this research is to investigate the impact of AI on intrinsic motivation and learning in different majors, academic levels, and cultural settings to advance global education. Studying AI-driven educational tools and their impact on students' intrinsic motivation provides valuable insights into effective teaching methods. In this study, educators and policymakers gain valuable insights into how to integrate and utilize AI technology in education. By analyzing students' opinions on AI in education, this study provides important perspectives on how AI affects educational practices and future research efforts.

Literature Review

AI and Education

AI technology has greatly transformed contemporary teaching. AI refers to the extensive domain of automated decision-making without human involvement (Tapalova & Zhiyenbayeva, 2022). Generative AI (GAI) technologies, such as machine learning, have enhanced digital content generation (Baidoo-Anu & Ansah, 2023). GAI is a machine learning framework that utilizes statistical approaches and probability to generate artificial artifacts, with or without supervision (Baidoo-Anu & Ansah, 2023; Jovanovic & Campbell, 2022; Ooi et al., 2023). GAI utilizes deep learning to create artificial objects from video, photos, text, and voice. This approach examines training instances for patterns and dispersion (Mannuru et al., 2023). Two primary generative AI models, generative pretrained transformers and generative adversarial networks, were identified in the works of Baidoo-Anu and Ansah (2023), Brown et al. (2020) (2020), and Jovanović and Campbell (2022). Generative adversarial networks are the most often used methods for developing artificial intelligence and utilize discriminator and generator neural networks.

Generative AI models have demonstrated remarkable proficiency in many natural language processing tasks, such as language comprehension, logic, and language creation (Govender, 2024). The AI community is currently questioning the capabilities and limitations of these models, and it is evident that assessing generative AI is quite difficult (Ahuja et al., 2023; Wang et al., 2023). Generative pre-trained transformer (GPT) models comprehend and generate multilingual text resembling human language by utilizing a vast amount of publicly accessible digital content data (Ahuja et al., 2023). Aydin and Karaarslan (2023) showed that these models are capable of generating compelling research paper paragraphs on various subjects. The models are

capable of simulating human interactions, such as chatbots or video game characters (Jovanovic & Campbell, 2022; Pavlik, 2023).

The 175 billion parameters of GPT-3 enhance performance in different tasks and compete with the latest fine-tuning methods (Brown et al., 2020; Kublik & Saboo, 2022; Motlagh et al., 2023). According to Brown et al. (2020), GPT-3 outperforms all nonsparse language models by a margin of 10. The GPT-3 language model feature is utilized in various sectors, such as education, engineering, journalism, medicine, economics, and finance.

Numerous studies have investigated the effects of using AI-driven tools in higher education. Strzelecki and ElArabawy (2024) used the unified theory of adoption and use of technology to analyze ChatGPT usage to investigate how AI impacts higher education. They examined the ways in which variables such as effort expectancy, performance expectation, social impact, and enabling variables affect students' behavior. The results showed that behavioral intentions are highly influenced by performance, effort, and social influence. Usage is influenced by behavioral intention and supportive factors. This study improved our understanding of how AI tools are being used in higher education. Customized training, user-friendly technology, and awareness campaigns enhance student attitudes and the use of AI tools in the classroom. The same approach was used to investigate how AI impacts higher education teachers (Strzelecki et al., 2024).

Additionally, Berendt et al. (2020) examined the potential drawbacks and advantages of AI in the field of education. They investigated the potential of AI and 'big data' to enhance real-time surveillance of the education system and its implications for the fundamental human rights and freedoms of teachers and students. The analysis indicated that creating, promoting, and implementing AI products involves managing advantages and risks. They concluded by advocating for the advancement, promotion, and implementation of AI educational technology solutions to assess its advantages and drawbacks. Artificial intelligence impacts data protection, privacy, and fundamental rights.

Intrinsic Motivation and AI

Intrinsic motivation is the behavior of engaging in an activity because of real pleasure or curiosity, as described by Ryan and Deci (2022). The fundamental principle of self-determination is intrinsic motivation. Extrinsic rewards, or the "carrot and stick" approach to motivation, along with deadlines, assessments, and strict supervision, have been shown to reduce people's intrinsic motivation for an activity, while giving them choices and recognizing their emotions tends to increase it (Douds, 2022; Ryan & Deci, 2022).

Intrinsically motivated individuals participate in personal enjoyment or challenges rather than in external rewards or coercion. Students with intrinsic motivation usually find pleasure in their tasks and demonstrate a spontaneous preference for them. Innate drive boosts creativity, engagement, and independence. Students might meet their psychological needs through activities motivated by internal factors. Technology has enabled internal motivational activities such as online classroom discussions to boost students' confidence and decision-making, team-based workshops to

improve teamwork, and collaborative online programs to reduce technical anxiety and enhance self-efficacy (Lin & Wang, 2021).

Scholars are focusing on the connection between AI and intrinsic motivation in the process of learning languages. They indicate that AI-driven tools, which provide customized feedback, incorporate game-like features, and provide interactive material, can boost learners' intrinsic motivation. Moreover, learners who utilized AI-powered language applications showed increased engagement and maintained interest (Chichekian & Benteux, 2022; Wang et al., 2024).

Lee et al. (2022) suggested that using an AI-powered chatbot can provide students with immediate and high-quality feedback for postclass evaluation. They utilized an AI-powered chatbot to assess students' receptiveness to learning, mentality, and academic achievements. Students from two universities participated. Eighteen experimental students and 20 control students participated in the same lesson. One group utilized an AI-based robot for postclass assessment, whereas the other group applied conventional methods. Researchers have discovered that employing AI-driven robots to evaluate public health courses can enhance students' academic achievement, boost their self-assurance, improve their eagerness to learn, and have a favorable influence on their attitude toward education. Chatbots have the potential to enhance student engagement in the educational process. Providing students with ample feedback during the review process can help them feel appreciated and foster a favorable rapport, ultimately resulting in improved academic performance.

AI progress will alter educational procedures. Chiu et al. (2023) discovered a correlation between AI and educational viewpoints or teaching methods, especially in K-12 education. AI can be beneficial for motivated and skilled students. It is essential to comprehend how instructors utilize student motivation to enhance and promote AI learning. This study utilized self-determination theory to investigate how teacher aid impacts the connection between student expertise, needs fulfillment, and intrinsic motivation to learn using technology. Chatbots were utilized in this AI-driven research involving 123 students. Teacher assistance and student skills, such as self-regulated learning and digital literacy, influenced intrinsic motivation and chatbot learning ability. Teacher support improved connection but reduced independence.

AI and Enhancing Learning

AI tools are increasingly being used in education to improve learning and teaching methods (T. Wang & Cheng, 2021). However, there is still an unresolved issue about the openness of the created data. Typically, when AI technologies are utilized to involve users in a specific learning task, users are expected to provide consent, indicating their agreement with the terms of technology use and the collection, management, and processing of their data (Nguyen et al., 2023).

AI progress in education can revolutionize the educational environment and impact the role of all stakeholders. Recently, AI technologies have been increasingly used to improve our comprehension of students' learning and to increase learning outcomes and experiences. The implementation of AI has resulted in increasing ethical hazards and worries related to several areas, such as personal data and learner independence (Nguyen et al., 2023). In several ways, AI has revolutionized formal

education. With artificial intelligence, students' performance data are analyzed and adapted to their individual needs. As a result, learning has become more efficient and effective. Teachers and students benefit from intelligent tutoring systems powered by artificial intelligence, which provides immediate feedback and guidance. With AI, adaptive assessments are created that adapt to students' performance. Teachers may concentrate more on teaching by using AI to automate administrative duties such as attendance scheduling and grading. Teachers can keep an eye on their students' progress, offer criticism, and respond to their inquiries with the help of AI-powered assistants. AI generates practice questions, summaries, and quizzes as well as personalized learning materials. The learning process is ongoing and lifelong since AI monitors students' progress and modifies learning paths accordingly (Alam, 2023).

Alam (Alam, 2023) investigated how AI and other contemporary technologies are enhancing student education and being incorporated into curricula and classrooms. The researcher mentioned that students can adapt learning experiences to their needs and skill levels and provide instant feedback. Furthermore, allowing interaction with virtual settings and simulations may provide students with immersive learning experiences. On online learning platforms, students can enroll in courses worldwide and take advantage of educational resources, as well as technologically enhanced teaching strategies tailored to their learning styles, goals, and speeds. The use of game-like elements makes studying more participatory and interesting for students. Moreover, AI has changed traditional education.

Klayklung et al. (2023) investigated the advantages of ChatGPT. They argue that conversational AI has enhanced learning in education. With ChatGPT's personalized and engaging conversations, education can be revolutionized. They found that ChatGPT can tailor educational experiences according to students' preferences and requirements. Furthermore, it enables students to engage meaningfully with plain language as well as comprehend it. Teachers can also use ChatGPT to provide students with more meaningful and personalized interactions by using ChatGPT as a tool for grading and providing feedback. ChatGPT also provides students with support and instructional materials outside of the classroom.

Methodology

A comprehensive methodological framework is provided to investigate the relationships between AI, intrinsic motivation, and learning enhancement in education. A nonquantitative and nonexperimental design was used through surveys, with participants from four countries (Saudi Arabia, Egypt, Spain, and Poland) pursuing varying levels and majors of faculty of education. A cross-sectional design is one of the most basic tools for conducting research. In addition to being inexpensive to perform, it can save participants' and researchers' time and offer sufficient responses to a wide range of queries. The research question can be approached in a way that can be developed as additional information becomes available that can serve as a foundation for future study designs (Spector, 2019).

Google Forms was used to gather data over a three-month period. Prior to the start of data collection, ethical approval was obtained to uphold ethical standards and

secure the rights of participants. All participants provided informed consent, indicating that they were aware of the purpose and methodology of the study and that they were willing to participate. The study's ethical integrity was ensured, and participants' rights and confidentiality were secured through consent and ethical approval processes.

Participants

This study used convenience sampling to select its sample; participants were chosen based on their availability and desire to participate (Simkus, 2023). A total of 455 students responded to the survey in the first quarter of 2024, 57.80% of whom were female ($n=263$) and 42.20% of whom were male ($n=192$). In terms of majors, English as a Foreign Language accounted for 28.1% of the respondents, followed by Primary Education (26.6%). Academic levels demonstrate a diverse distribution across freshmen (27.9%), sophomores (14.1%), juniors (22.9%), and seniors (35.2%). Nationality-wise, Egyptian respondents constituted the largest group at 31.9%, followed by Saudi (25.5%), Spanish (24.6%), and Polish (18.0%) respondents.

Regarding to the gender, in Spain, female students predominate at all academic levels, especially in Senior (85.7%) compared to males (14.30). In Saudi Arabia, there is a female majority in Freshman, Junior and Senior, but in Sophomore, male students dominate (71.4%). In Egypt, the distribution is more balanced, with an equal gender proportion in Senior and a female predominance in the Junior specialty (71.4%). And finally, in Poland, male students are the majority at all levels, being most notable in the Freshman specialty (80%). In general, the gender distribution varies considerably between territories and academic levels, with a greater female presence in Spanish and Saudi Arabia, a balanced distribution in Egyptian, and a male predominance in Polish. Regarding the age of students, the age distribution shows that in Spain women predominate at all academic levels and have a slightly lower average age compared to men. In Saudi Arabia, although women are the majority at most levels, men dominate in Sophomore and the average age varies without a clear pattern. In Egypt, the distribution is more balanced with similar average ages between genders. In Poland, men are the majority at all levels and have average ages close to but slightly lower than those of women (Table 1). These demographic profiles provide an essential context for understanding the respondent population and can inform future research and decision-making. This strategy made a mixture of diverse, geographically dispersed students from different cultural backgrounds possible.

Instrumentation

To carry out the purposes of the study, the authors developed an instrument that measures the potential of AI to empower students. The initial version of the instrument was composed of a total of 21 items classified into three dimensions. The first dimension (DIM. A) was titled "Intrinsic motivation and AI", with a total of seven items. The second dimension (DIM. B) was titled "AI, and enhancing learning", with a total of 7 items; and the third dimension (DIM. C) was titled "Overall perception of AI in education", with a total of 7 items. A 5-point Likert scale was used to measure

Table 1 Demographic and seniority distribution

Country	Seniority	Female		Male	
		N (%)	Average age	N (%)	Average age
Spanish	Freshman	72.4%	18.73	27.6%	19.26
	Sophomore	50.0%	19.00	50.0%	20.00
	Junior	80.0%	18.25	20.0%	24.00
	Senior	85.7%	19.33	14.3%	26.00
Saudi	Freshman	72.7%	18.75	27.3%	18.67
	Sophomore	28.6%	19.25	71.4%	20.20
	Junior	65.0%	20.85	35.0%	22.71
	Senior	63.4%	22.60	36.6%	21.85
Egyptian	Freshman	53.8%	20.71	46.2%	17.50
	Sophomore	53.8%	18.86	46.2%	19.00
	Junior	71.4%	20.40	28.6%	20.29
	Senior	50.0%	22.50	50.0%	26.05
Polish	Freshman	20.0%	24.00	80.0%	22.25
	Sophomore	33.3%	20.33	66.7%	22.33
	Junior	23.3%	23.14	76.7%	22.91
	Senior	39.5%	24.27	60.5%	23.52

the items. For the DIM. A and DIM. B the value of 1 was associated with a strongly disagree label, while a value of 5 was associated with a strongly agree label. For the DIM. C, the value 1 was associated with very negative, while the value 5 was associated with very positive.

First, the authors validated the instrument (construct) through exploratory factor analysis and confirmatory factor analysis. This validation was achieved after several model attempts, with a final version of 10 items (DIM. A, 3 items; DIM. B, 3 items; and DIM. C, 4 items).

Exploratory factor analysis was conducted using the Varimax orthogonal rotation method, allowing the factors to be related (Basto & Pereira, 2012). Items with factor loadings lower than 0.40 were eliminated as a criterion to guarantee adequate saturation in their respective factors (Sánchez et al., 2019). The results indicated a Kaiser–Meyer–Olkin (KMO) index of 0.919, suggesting an adequate sample for factor analysis, and Bartlett’s test of sphericity was significant ($\chi^2 = 2312.330$; $df = 455$; $p < 0.05$), which supports the suitability of the data. The final version of the factor model explained 71.56% of the true variance in the scores provided by participants. Specifically, dimension 1 explained 6.57% of the variance, dimension 2 explained 54.70%, and factor 3 explained 9.88%. This finding suggests a robust and well-defined factor structure in the instrument used.

Afterwards, confirmatory factor analysis was carried out to evaluate the discrepancy between the observed data and those predicted by the model using both correlation and covariance matrices. For this purpose, the maximum likelihood method was applied, which assumes that the items exhibit a multivariate normal distribution. The validity of this assumption was evaluated using the Mardia coefficient, which is considered acceptable if its value is lower than the result of the formula $p(p+2)$ (Bollen, 1989), where p represents the number of items in the factor model (in our case, 10 items in the final version). In this case, a value of 78.614 was obtained, which suggests the normality of the matrix.

To evaluate the adequacy of the model, several indices were considered. The first is the CMIN/df (minimum discrepancy/degrees of freedom), where a value ≤ 3 indicates an acceptable fit (Kline, 2011). In this study, the result obtained in AMOS was 2,590. Regarding the error rates, the comparative fit index (CFI) was 0.978, and the normed fit index (NFI) was 0.964. These values, ranging from 1 to >0.95 , suggest a perfect fit to the model (Mia et al., 2019). For the root mean square error of approximation (RMSEA) index, a value between 0.05 and 0.08 indicates a reasonable approximation error (MacCallum et al., 1996). In this study, the AMOS result was 0.059, within the acceptable range. Finally, the incremental fit index (IFI) and Tucker–Lewis coefficient (TLI) are incremental fit indices. Values close to 1 indicate a very good fit (Whittaker & Schumacker, 2022). In this analysis in AMOS, coefficients of IFI=0.978 and TLI=0.969 were obtained, indicating a very good fit.

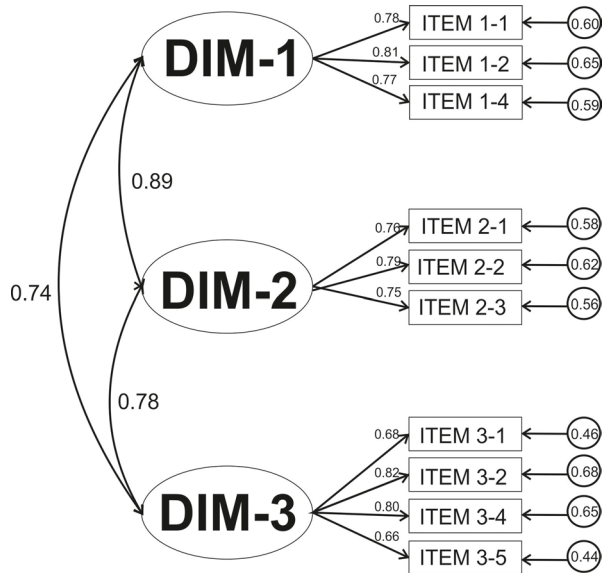
Regarding discriminant validity, Cheung and Wang (2017) establish that there is validity if the values of the average variance extracted exceed 0.50 (according to the Fornell–Larcker criterion): DIM.1=0.615; DIM. 2=0.584; DIM. 3=0.556. Several reliability coefficients (Cronbach’s alpha and composite reliability) were applied in connection to the instrument’s internal consistency. It is recommended by the literature that the coefficients be larger than 0.7 (Taber, 2018). Regarding the first, the coefficients were the following: DIM.1=0.828; DIM. 2=0.806; DIM. 3=0.824, while for the second index, DIM. 1=0.827; DIM. 2=0.808; DIM. 3=0.832. Both indices were satisfactory. The final version of the instrument is shown in Table 2, and the structural equation model is shown in Fig. 1.

The reliability and internal consistency of the data collection instrument were assessed using Cronbach’s alpha coefficient and correlation analysis. The calculated Cronbach’s alpha value of 0.907, which falls within the recommended range of 0.7 to 1.0, indicates strong internal consistency among the questionnaire items, confirming the reliability of the instrument for data analysis. Correlation analysis further corroborated the internal consistency validity, with all correlation coefficients between individual statements and the total score demonstrating positive and statistically sig-

Table 2 Final version of the instrument

DIM. 1 Intrinsic Motivation and AI
1.1 AI-driven virtual environments have enhanced my learning experience.
1.2 In my major, AI-based tools have enabled me to develop autonomy and self-direction.
1.3 AI-driven gamification has improved my Intrinsic motivation to engage in major activities.
DIM. 2 AI, and Enhancing Learning
2.1 I have been able to focus on areas where I need improvement using AI systems.
2.2 Using AI technologies has enhanced my critical thinking and problem-solving skills.
2.3 AI-driven tools have provided me with opportunities for personalized and tailored learning experiences in my major.
DIM 3 Overall Perception of AI in Education
3.1 How would you rate the impact of AI technologies on the education landscape as a whole?
3.2 To what extent do you believe AI can contribute to the continuous improvement of your major?
3.3 In your opinion, how well does AI align with the evolving needs and preferences of learners in your major?
3.4 How confident are you in the ethical use of AI in your major education within the Faculty of Education to safeguard learner privacy and security?

Fig. 1 Structural equation modeling



nificant associations at the 0.01 level. Additionally, correlations between dimensions such as intrinsic motivation and AI, between AI and enhancing learning, and between overall perception of AI in education also exhibited strong positive associations with the total score. These findings underscore the stability and suitability of the results for the study, reinforcing the reliability and validity of the data collection instrument in estimating respondents' perceptions and experiences regarding AI technologies in education.

The correlation coefficients between each statement's score and the total score for all items are displayed in Table 3. At the 0.01 level, every correlation coefficient that existed between each item's score and the overall score was positive and statistically significant. The correlation coefficients between each dimension's score and the overall score for all items are displayed in Table 4. At the 0.01 level, every correlation coefficient that existed between each item's score and the overall score was positive and statistically significant.

Data Analysis

This study utilized various statistical methods to examine the impact of AI-driven language learning aids on students' intrinsic motivation and learning outcomes across diverse academic disciplines, educational levels, and cultural backgrounds. SPSS 26 was used for the data analysis. Initially, descriptive statistics were used to understand the general patterns in the data. The analysis involved computing averages, standard deviations, standard errors, and confidence intervals to characterize students' perceptions of AI-driven instructional aids. Analyses of variance (ANOVA) were performed to assess variations in intrinsic motivation across various student groups. The groups were compared according to nationality, major, and academic standing. Significant

Table 3 Internal consistency validity

Correlations		
DIM 1 (N=455)	Pearson Correlation	Sig. (2-tailed)
1. AI-driven virtual environments have enhanced my learning experience.	0.755**	0.000
2. In my major, AI-based tools have enabled me to develop autonomy and self-direction.	0.771**	0.000
3. AI-driven gamification has improved my Intrinsic motivation to engage in major activities.	0.733**	0.000
DIM 2(N=455)		
4. I have been able to focus on areas where I need improvement using AI systems.	0.745**	0.000
5. Using AI technologies has enhanced my critical thinking and problem-solving skills.	0.754**	0.000
6. AI-driven tools have provided me with opportunities for personalized and tailored learning experiences in my major.	0.769**	0.000
DIM 3(N=455)		
7. How would you rate the impact of AI technologies on the education landscape as a whole?	0.697**	0.000
8. To what extent do you believe AI can contribute to the continuous improvement of your major?	0.752**	0.000
9. In your opinion, how well does AI align with the evolving needs and preferences of learners in your major?	0.740**	0.000
10. How confident are you in the ethical use of AI in your major education within the Faculty of Education to safeguard learner privacy and security?	0.671**	0.000

Table 4 The correlation coefficients between the score of each dimension

Correlations			
	N	Pearson Correlation	Sig. (2-tailed)
Intrinsic Motivation and AI	455	0.873**	0.000
AI and Enhancing Learning	455	0.890**	0.000
Overall Perception of AI in Education	455	0.879**	0.000

differences shown by ANOVA were further analyzed using post hoc tests, including Tukey's HSD test and Scheffé's test, to identify specific group differences.

Results

Answering RQ1: “What is the impact of AI in education on intrinsic motivation and enhancing learning across different majors, levels, and cultures?” The following results were estimated. Table 5 displays the ANOVA results for the variable “nationality”, showing a statistically significant variation in intrinsic motivation among the various groups, with a p value of 0.011, which is below the 0.05 threshold. Intrinsic motivation differs greatly depending on students' nationality. The multiple comparisons table displays the outcomes of the LSD post hoc test, indicating the distinct groups that exhibit differences from one another. Polish pupils demonstrate the high-

Table 5 ANOVA: intrinsic motivation and AI by nationality, Major, and academic level

Intrinsic Motivation and AI (Nationality)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	72.554	3	24.185	3.728	0.011
Within Groups	2925.477	451	6.487		
Total	2998.031	454			
Intrinsic Motivation and AI (Major)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	182.386	4	45.596	7.287	0.000
Within Groups	2715.683	434	6.257		
Total	2898.068	438			
Intrinsic Motivation and AI (Academic Level)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	43.884	3	14.628	2.233	0.084
Within Groups	2954.147	451	6.550		
Total	2998.031	454			

est levels of intrinsic motivation, followed by Egyptian students. Spanish and Saudi students rank lower in that order. The results highlight how cultural and national origins influence students' intrinsic motivation when using AI-powered language learning tools.

Additionally, the ANOVA for the variable “major” shows a significant difference across group means, with a p-value of 0.000, suggesting statistical significance below the 0.05 level. Students' majors have a substantial impact on their intrinsic motivation levels. The multiple comparisons table displays the outcomes of the LSD post hoc test, highlighting the particular groups that exhibited significant differences. Students majoring in education technology exhibit the greatest intrinsic drive, followed by those in special education and English as a foreign language. Primary education and early childhood majors have decreasing levels of intrinsic motivation in that sequence. The results demonstrate the differing levels of inherent motivation in various academic disciplines, underscoring the impact of academic focus on how students interact with AI-based educational resources.

In contrast, the ANOVA for the variable “academic level” showed no statistically significant difference in group means, with a significance value of $p=0.084$, which is above the 0.05 threshold. This indicates that intrinsic motivation does not significantly differ based on students' academic level, whether they are freshmen, sophomores, juniors, or seniors. Thus, intrinsic motivation influenced by AI-powered learning tools appears to be consistent across different academic levels, suggesting that factors other than academic standing play a more crucial role in determining students' motivation.

Table 6 displays the ANOVA results for the variables “nationality, major, and academic level”, showing a statistically significant variation in Enhancing Learning among the various groups. In Table 6, an ANOVA analysis was performed to examine the effects of nationality, major, and academic level on AI and Enhancing Learning. There was a statistically significant difference between different nationalities

Table 6 ANOVA: enhancing learning and AI(nationality, major, academic level)

AI and Enhancing Learning (Nationality)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	7.949	3	2.650	3.535	0.015
Within Groups	338.019	451	0.749		
Total	345.967	454			
AI and Enhancing Learning (Major)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	16.574	4	4.143	5.700	0.000
Within Groups	315.508	434	0.727		
Total	332.081	438			
AI and Enhancing Learning (Academic Level)					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	3.698	3	1.233	1.624	0.183
Within Groups	342.269	451	0.759		
Total	345.967	454			

Table 7 The mean and standard deviation of the scale dimensions

	N	Mean	Std. Deviation
Intrinsic Motivation and AI	455	10.6462	2.56975
Enhancing Learning and AI	455	10.6330	2.61885
Overall Perception of AI in Education	455	14.1978	3.18525
Valid N (listwise)	439		

for the variable “Nationality” ($p=0.015$), suggesting that nationality is an important factor in AI and Enhancing Learning. Additionally, for the variable “Major”, there was a significant difference ($p=0.001$), indicating that major choices significantly impact AI and enhancing learning. Academic level, however, was not statistically significant ($p=0.183$), suggesting academic level does not significantly influence AI performance. The findings indicate that it is important to consider both nationality and major when analyzing the effects of AI and Enhancing Learning, while academic level may not be a significant factor at this time.

Table 7 presents the descriptive statistics for the scale dimensions related to AI in education. The table shows the mean scores and standard deviations for three key areas: intrinsic motivation influenced by AI, the enhancement of learning through AI, and the overall perception of AI in education. The mean scores for intrinsic motivation and enhancing learning are 10.6462 and 10.6330, respectively, with standard deviations of approximately 2.57 and 2.62, indicating moderate levels of variation among respondents. The overall perception of AI in education had a higher mean score of 14.1978 and a standard deviation of 3.19, suggesting a generally positive view with a slightly wider range of opinions. The sample size for each dimension is 455, with a valid listwise sample size of 439.

This study examined the impact of AI-powered learning tools on intrinsic motivation among students. The second research question focused on understanding these

effects more deeply. Answering RQ2: “How do AI-powered learning tools affect students’ intrinsic motivation?” The following results were achieved. Table 8 shows that the participants expressed that AI-driven virtual environments greatly improved their learning experience, with an average score of 3.6. Students found these virtual environments to be advantageous for their learning efforts. The mean’s 95% confidence interval spanned from 3.5108 to 3.6892, indicating a range of values where the true population mean is likely to lie. The 5% trimmed mean and median scores indicate a sense of stability and an overall positive perception of AI-driven virtual environments. The scores were 3.6453 and 4.0, respectively. It is worth mentioning that the standard deviation of 0.96776 suggests a certain level of variability in the responses.

In addition, participants expressed that AI-based tools for autonomy and self-direction in their major subjects had a significant impact on their intrinsic motivation, scoring an average of 3.5407. The mean 95% confidence intervals ranged from 3.4476 to 3.6337. It appears that students found the use of AI-based tools to be beneficial for fostering their autonomy and self-direction in learning. The 5% trimmed mean and median scores were 3.5769 and 4.0, respectively, suggesting a generally positive overall perception. Nevertheless, the standard deviation of 1.00986 indicates

Table 8 The effects of AI-powered learning tools on students’ intrinsic motivation

			Statistic	Std. Error	Rank
AI-driven virtual environments have enhanced my learning experience.	Mean		3.6000	0.04537	1.000
	95% Confidence Interval for Mean	Lower Bound	3.5108		
		Upper Bound	3.6892		
	5% Trimmed Mean		3.6453		
	Median		4.0000		
	Variance		0.937		
	Std. Deviation		0.96776		
In my major, AI-based tools have enabled me to develop autonomy and self-direction.	Mean		3.5407	0.04734	2.000
	95% Confidence Interval for Mean	Lower Bound	3.4476		
		Upper Bound	3.6337		
	5% Trimmed Mean		3.5769		
	Median		4.0000		
	Variance		1.020		
	Std. Deviation		1.00986		
AI-driven gamification has improved my Intrinsic motivation to engage in major activities.	Mean		3.5055	0.04697	3.000
	95% Confidence Interval for Mean	Lower Bound	3.4132		
		Upper Bound	3.5978		
	5% Trimmed Mean		3.5403		
	Median		4.0000		
	Variance		1.004		
	Std. Deviation		1.00191		

a certain level of variability in the responses, highlighting the importance of taking individual differences into account.

In addition, participants expressed that the implementation of AI-driven gamification had a positive impact on their internal drive to participate in significant activities, as indicated by an average score of 3.5055. The mean 95% confidence intervals ranged from 3.4132 to 3.5978. Students found that the implementation of gamified elements powered by AI had a positive impact on their motivation. The 5% trimmed mean and median scores were 3.5403 and 4.0, respectively, indicating a generally positive perception. Like in the previous statements, there was some variability in the responses, as indicated by the standard deviation of 1.00191 points.

The study's third research question, "To what extent does AI integration enhance learning?", investigated how AI integration impacts learning. The results in Table 9 reveal the effects of AI integration on learning. The participants indicated that AI systems helped them concentrate on areas needing improvement, with an average score of 3.6330. This shows that AI technologies have enabled students to recognize and work on their specific learning requirements. The standard deviation of 1.01715 indicates varying responses, highlighting individual differences in the effectiveness of AI systems for focusing on improvement.

Furthermore, participants reported that using AI technologies enhanced their critical thinking and problem-solving skills, with a mean score of 3.4703. This finding suggested that AI integration in learning contexts positively influenced participants' higher-order thinking abilities. The standard deviation of 1.06955 indicates some variability in the reported enhancement of critical thinking and problem-solving skills.

Moreover, participants indicated that AI-driven tools provided them with opportunities for personalized and tailored learning experiences in their major subjects, with a mean score of 3.5297. This implies that AI integration has enabled students to engage in learning activities that are specifically tailored to their individual needs and interests. The standard deviation of 0.99708 suggests some variability in the perceived level of personalization and tailoring facilitated by AI-driven tools.

The descriptive statistics provide valuable insights into the participants' perceptions of the extent to which AI integration enhances learning. The mean scores indicate a moderate level of agreement with the positive effects of AI systems on focusing on improvement, enhancing critical thinking and problem-solving skills, and providing personalized learning experiences. However, the standard deviations highlight the variability in participants' experiences and perceptions.

Table 9 AI and enhancing learning

	<i>N</i>	Mean	Std. Deviation	Rank
I have been able to focus on areas where I need improvement using AI systems.	455	3.6330	1.01715	1.000
Using AI technologies has enhanced my critical thinking and problem-solving skills.	455	3.4703	1.06955	3.000
AI-driven tools have provided me with opportunities for personalized and tailored learning experiences in my major.	455	3.5297	0.99708	2.000
Valid N (listwise)	455			

RQ4, “What are students’ overall perceptions of the impact of AI technologies on education?”, aimed to examine the overall perception of students regarding the impact of AI technologies on education. The results shown in Table 10 provide insights into these perceptions. Participants rated the impact of AI technologies on the education landscape as a whole with a mean score of 3.6264. This suggests that, on average, students perceive the impact of AI technologies on education to be moderately positive. The standard deviation of 0.94622 indicates some variability in the ratings, reflecting different perspectives among the participants.

Furthermore, participants expressed their belief in the extent to which AI can contribute to the continuous improvement of their major, with a mean score of 3.6484. This indicates that students hold a moderate belief in the potential of AI to contribute to the continuous improvement of their major. The standard deviation of 0.93363 suggested some variability in the level of belief among the participants. Moreover, participants provided their opinions on how well they believe AI aligns with the evolving needs and preferences of learners in their major, with a mean score of 3.5978. This implies that students perceive a moderate alignment between AI technologies and the evolving needs and preferences of learners in their major. The standard deviation of 0.94029 suggested some variability in the perceived alignment among the participants. Additionally, participants indicated their confidence in the ethical use of AI in their major education to safeguard learner privacy and security, with a mean score of 3.3253. This suggests that students have a moderate level of confidence in the ethical use of AI for safeguarding learner privacy and security. The standard deviation of 1.10651 indicates some variability in the level of confidence among the participants.

Discussion

This [discussion](#) section explores the implications of the study’s findings, focusing on how AI-powered tools impact students’ intrinsic motivation and learning experiences. To gain insight into the cultural and contextual factors that influence the effectiveness of AI in education, we carefully examine the differences across nationalities, majors, and academic levels. Additionally, we explore students’ perceptions of AI technologies, highlighting their potential benefits and ethical considerations.

Table 10 The overall perception of students regarding the impact of AI technologies on education

	N	Mean	Std. Deviation
How would you rate the impact of AI technologies on the education landscape as a whole?	455	3.6264	0.94622
To what extent do you believe AI can contribute to the continuous improvement of your major?	455	3.6484	0.93363
In your opinion, how well does AI align with the evolving needs and preferences of learners in your major?	455	3.5978	0.94029
How confident are you in the ethical use of AI in your major education within the Faculty of Education to safeguard learner privacy and security?	455	3.3253	1.10651
Valid N (listwise)	455		

By providing justifications and identifying future research areas, we aim to provide meaningful context for these results within the literature.

The first research question reveals significant differences in intrinsic motivation based on nationality and major but not academic level. The highest levels of intrinsic motivation are observed among Polish students and those pursuing a major in education technology. This study confirms previous research emphasizing cultural context and academic discipline as important influences on educational motivation. The cultural factors affecting intrinsic motivation have been highlighted by Ryan and Deci (2022), who suggest that education practices and cultural values have a significant influence. In addition, Schunk and Zimmerman (2008) assert that self-regulation and autonomy enhance motivation within particular fields of study. Students' engagement and motivation with AI-powered language learning tools appear to vary significantly across nationalities, suggesting that cultural factors are important. These tools may be more motivating for students from cultures that place a high value on individual achievement and technological advancement.

In response to the second research question, AI-powered language learning tools are found to have a positive impact on students' intrinsic motivation to learn language. This finding aligns with earlier studies conducted by Deterding et al. (2011) and Hamari et al. (2014), which found that gamification and interactive learning environments significantly enhance motivation and engagement. Students perceive AI-driven virtual environments, tools that promote autonomy and self-direction, and gamification elements as enhancing their learning experience and motivation to actively participate in their studies. The positive influence can be attributed to AI tools' ability to provide immediate feedback, create engaging and personalized learning experiences, and incorporate gamified elements that foster a sense of achievement. These features make the learning process more enjoyable and relevant to students' individual interests and needs.

Regarding the third research question, the study suggests that integrating AI in education can enhance learning outcomes by allowing students to focus on areas requiring improvement, fostering critical thinking and problem-solving skills, and providing personalized learning experiences. These findings are supported by research conducted by Huang and Rust (2018), who identified personalized learning as a crucial factor in improving educational outcomes. The adaptability of AI to individual learning styles and its capacity to offer customized support help address each student's specific needs, thereby enhancing their overall learning experience. However, the existence of individual differences among students highlights the need for further research to identify the specific conditions and contexts in which AI technologies can most effectively enhance learning outcomes. This emphasizes the importance of developing AI applications that are flexible and capable of catering to diverse learning preferences and needs.

According to the fourth research question, students perceive AI technologies as having a moderately positive impact on education. There is a reasonable alignment between AI technologies and evolving learner needs and preferences, and they believe that AI contributes to the continuous improvement of their respective majors. These perceptions are consistent with findings from studies conducted by Luckin et al. (2016) and Mohamed (2024), which indicate that while AI is generally viewed

positively, concerns regarding its implementation and ethical use exist. Moreover, students express a moderate level of confidence in the ethical use of AI to safeguard learner privacy and security, confirming concerns raised by scholars such as Binns (2018) and Alenezi et al. (2023). The findings emphasize the importance of taking into account students' perspectives and addressing ethical concerns when integrating AI technologies into education. In addition to being effective, AI applications must be trusted and accepted by learners to be successfully adopted and improve education quality.

Conclusion

This study demonstrates AI-powered language learning aids might boost intrinsic motivation, especially in certain nationalities and majors. AI's favourable effects on motivation and learning highlight the need for deliberate integration in education. The results are intriguing, but further research is needed to understand the long-term consequences and improve AI tools for different learners. Addressing ethical problems and embracing students' opinions would help AI in education succeed, improving learning experiences.

This study contains limitations. First, convenience sampling may not fully represent the student population, limiting generalizability. Second, the survey's cross-sectional design may not represent long-term motivation or perception changes. Social desirability and erroneous self-assessment can skew self-reported data, which the study used. The study only included a few nations and majors, which may have overlooked cultural and academic aspects that affected the outcomes.

To overcome these limitations, future research should use a more representative sampling strategy and a longitudinal design to track improvements. Expanding the study to include other countries and academic disciplines would help explain AI's effect on intrinsic motivation. By examining students' perspectives, qualitative studies can supplement quantitative data. Investigating AI tools' motivation and learning-boosting qualities could help design more tailored and effective educational solutions.

The findings from this study have significant pedagogical consequences. The study demonstrates the revolutionary potential of AI-driven pedagogical resources. Specifically, when AI is used effectively, it can enhance student autonomy, critical thinking, and enhancing learning. To maximize their usefulness in the classroom, AI systems should be able to adapt to the culture and subject matter.

However, trust among students is a critical issue. In order to gain students' trust, AI ethics must be given top priority. Problems with algorithmic transparency, data security, and privacy must be resolved before these technologies can reach their revolutionary potential. This study also suggests that incorporating AI tools into the classroom calls for substantial professional development and training for educators. Learning responsible usage of apps powered by AI is a great way for teachers to engage and inspire their students. In order to help teachers make the most of these new learning opportunities, AI training should be a part of teacher preparation programs. Hence, the study can assist educators in implementing AI technology in a way that prioritizes student autonomy, critical thinking, customized learning, and ethics.

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Declarations

Ethical Approval We hereby declare that this manuscript is the result of our independent creation under the reviewers' comments. Except for the quoted contents, this manuscript does not contain any research achievements that have been published or written by other individuals or groups. We are the only authors of this manuscript. The legal responsibility of this statement shall be borne by us.

Competing Interests The authors have declared no competing interests.

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